Strict versus Permissive Reasoning about Disconfirming Evidence

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Introduction

Instant vs. Delayed Rationality

As they conduct research, scientists evaluate hypotheses against evidence, rejecting those that are disconfirmed. This *instant rationality* propels empirical science. However, this works less for theoretical science, where *delayed rationality* – tolerating disconfirming evidence – is often important. This study investigated differences in how middle school children reason about disconfirming evidence.

Strict vs. Permissive Reasoners

We hypothesized that some children are naturally inclined towards *instant rationality.* These *strict reasoners* prefer instant rationality, and quickly dismiss hypotheses that have been disconfirmed. We hypothesized that other children are *permissive reasoners*: They prefer delaying rationality, and are willing to retain hypotheses in the face of disconfirming evidence for further evaluation.

Hindsight Bias & Confirmation Bias

We developed behavioral measures of strict versus permissive reasoning about disconfirming evidence by recasting two constructs from the judgment and heuristics literature. *Hindsight bias* is the tendency for people to silently revise their beliefs upon seeing negative evidence, and to later claim they "knew it all along" (Bernstein, Atance, Meltzoff, & Loftus, 2007; Fischoff, 1975; Hawkins & Hastie, 1990). *Confirmation bias* is the tendency for people to look only for positive evidence for their beliefs, and to ignore negative evidence (Nickerson, 1998; Wason, 1960).

Research Focus

The current proposal is that both strict and permissive reasoners are rational, but that their rationality operates over different time scales (short versus long, respectively) and towards different goals (hypotheses evaluation versus theory development, respectively). The prediction, then, is that strict reasoners and permissive reasoners will not differ in overall cognitive ability, motivational orientation, or scientific reasoning, but rather in the relation between these factors.

Method

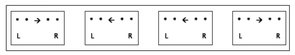
Participants & Design

Participants were 109 7th Grade Students (age: *M*=13.37, *SD*=0.29; 55 males, 54 females). As part of a correlational design, each participant completed all measures.

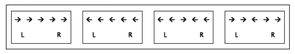
Measures

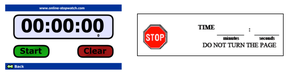
Measurement Target	
Response to Disconfirming Evidence	Science Quiz (Hindsight Bias) Coin Flip (Confirmation Bias)
Simple Cognitive Ability	Processing Speed Working Memory Selective Attention
Complex Cognitive Ability	Wisconsin Card Sort
Motivational Orientation	Implicit Theory of Intelligence Science Self-Efficacy
Scientific Reasoning	Five Facet Test of Scientific Reasoning

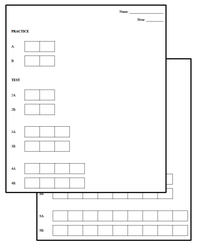
Conflict

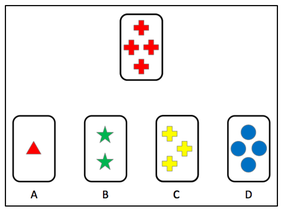


Non-Conflict

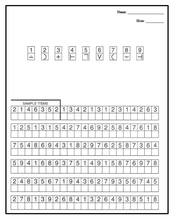








Whole-class version of WCST designed by Jeremy Wang

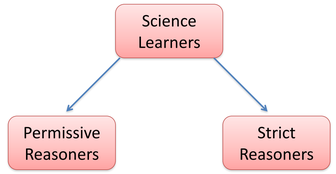


The scientific reasoning measure was administered by classroom teachers during one class period; participants required approximately twenty minutes to complete it. All other measures were administered by experimenters during another class period, and required approximately 40 minutes to complete.

Results

The Groups

The permissive group consisted of the 29 participants who demonstrated either hindsight bias or confirmation bias. The strict group consisted of the remaining 65 participants.



We first evaluated whether strict and permissive disconfirmers differ on cognitive ability, motivational orientation, or scientific reasoning, comparing the groups on each measure using independent *t*-tests. As predicted, there were no differences on any of the measures (p>0.30 for all).

Variable	Reasoning Group		*t*(92)	*p*
	Strict	Permissive		
Age	13.38 (0.28)	13.31 (0.29)	0.958[a]	0.341
Processing Speed	59.08 (11.87)	60.45 (13.68)	-0.493	0.623
WM	58.45 (11.32)	59.34 (11.13)	-0.357	0.722
Flanker	9.48 (9.16)	7.55 (6.32)	1.026	0.308
WCST Correct	36.38 (6.01)	35.41 (7.55)	0.666	0.507
WCST Systematic Errors	0.20 (0.78)	0.24 (0.91)	-0.836	0.405
WCST Perseveration Errors	0.71 (0.53)	0.86 (0.63)	-0.326	0.745
Implicit Intelligence	10.95 (4.00)	11.28 (3.65)	-0.369	0.713
Self-Efficacy	8.35 (2.92)	8.72 (2.64)	-0.584	0.561
Science Reasoning (*z*)	-0.015 (1.02)	0.017 (0.92)	-0.147	0.883

We next evaluated whether strict and permissive reasoners differ in the relation between cognitive ability and motivational orientation on one hand and scientific reasoning on the other. For each group, we regressed scientific reasoning on the cognitive and motivational variables, using a hierarchical approach to control for (0) age and to evaluate the additional predictive power provided by (1) basic cognitive ability, (2) complex cognitive ability, and (3) motivational orientation.

Step	Variable	Strict Reasoners			Permissive Reasoners		
		B (*z*)	*t*	*p*	B (*z*)	*t*	*p*
0	Control						
	(intercept)		-0.944	.349		-1.698	.106
	Age	0.083	0.732	.468	0.253	1.229	.233
1	Basic Cognition						
	Processing Speed	**0.378**	**2.752**	**.008**	**-0.564**	**-2.142**	**.045**
	WM	-0.175	-1.485	.144	0.580	2.835	.011
	Flanker	-0.070	-0.592	.556	0.243	1.413	.174
2	Complex Cognition (WCST)						
	Correct	-0.070	-0.611	.544	0.085	0.463	.648
	Systematic Errors	0.194	1.626	.110	**0.508**	**2.890**	**.009**
	Perseveration Errors	0.049	0.414	.680	**0.380**	**2.221**	**.039**
3	Motivation						
	Implicit Intelligence	**0.318**	**2.546**	**.014**	**0.742**	**2.826**	**.011**
	Self-Efficacy	-0.113	-0.785	.436	-0.262	-1.674	.111

Results & Conclusions

Strict Reasoners	Permissive Reasoners
Fast Processing Speed	Slow Processing Speed
	More Systematic Errors
	More Perseverative Errors



Strict Reasoners

As predicted, **better scientific reasoning** was associated with **faster processing**. One interpretation of this finding is that fluent evaluation of hypotheses against evidence frees up limited cognitive resources for higher-order inferencing.

Permissive Reasoners

As predicted, **better scientific reasoning** was associated with **slower processing** making **more systematic errors** on the WCST. We interpret these findings as evidence of a speed-accuracy trade-off. More deliberate permissive reasoners carefully accommodate hypotheses in the face of disconfirming evidence, whereas their more impulsive counterparts breeze by it without fully considering its implications, resulting in worse scientific reasoning.

Current Work

EEG Study

This research establishes a foundation for future research on the neural correlates of scientific reasoning. The differential behavioral findings for strict versus permissive reasoners make predictions about brain function. Currently, we are following up on these findings by investigating the areas of the brain that are involved as students engage in particular types of cognition and in scientific reasoning.

